

SENTHAMIZHVELAN M

Cloud & DevOps Engineer — AWS — Docker — Terraform — CI/CD

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PROFESSIONAL SUMMARY

Cloud & DevOps fresher with a B.Tech in Artificial Intelligence & Data Science and hands-on experience building and deploying AWS-based applications. Experienced with AWS, Linux, Docker, Terraform, GitHub Actions, CI/CD, monitoring, and infrastructure automation. Also possess practical knowledge of AI technologies including RAG, LLM APIs, Agentic AI, MCP, and workflow automation. Built and deployed end-to-end cloud projects on EC2, including an event-driven incident-response platform with automated remediation, secure CI/CD using GitHub OIDC, and SSM-based deployments.

TECHNICAL SKILLS

Cloud & AWS: EC2, VPC, IAM, S3, ECR, EBS, RDS, CloudWatch, EventBridge, API Gateway, SSM, SNS, SQS, Secrets Manager, ELB, Auto Scaling, ECS, EKS, Route 53, CloudFront, WAF

DevOps & CI/CD: Git, GitHub, GitHub Actions, Jenkins, Docker, Docker Compose, Kubernetes, Terraform, GHCR

Linux & Networking: Ubuntu Linux, Bash, SSH, Nginx, TCP/IP, DNS, HTTP/HTTPS, Subnets, Routing, Security Groups, Load Balancing

Monitoring & Security: Prometheus, Grafana, CloudWatch, CloudTrail, IAM Least Privilege, Health Checks, Secrets Management, Incident Response

Programming & Backend: Python, boto3, pytest, SQL

AI & Automation: RAG, Agentic AI, LLM APIs, MCP, n8n Workflow Automation

EXPERIENCE

Hema's Enterprises Private Limited (HEPL)

Feb 2026 – Aug 2026

AI Intern

- Automated internal business workflows using n8n, reducing manual processing time by **20%**.
- Built and deployed RAG-based internal knowledge retrieval systems integrated through REST APIs.
- Implemented Model Context Protocol (MCP) integrations to connect LLM workflows with internal tools and data sources.

PROJECTS

CloudGuardian – AI-Assisted CloudOps Incident Response Platform GitHub

AWS, Docker, GitHub Actions, Terraform, Python, FastAPI, Prometheus

- Designed an event-driven AWS incident-response workflow using **CloudWatch, EventBridge, API Gateway, FastAPI, Docker, EC2, and SSM** to process infrastructure alarms through detection, diagnosis, approval, remediation, and verification.
- Built a CI/CD pipeline with **GitHub Actions** integrating Ruff, pytest, and Trivy security scanning; successful builds publish Docker images to **Amazon ECR** and deploy to EC2 through AWS Systems Manager.
- Implemented **GitHub OIDC authentication** with short-lived AWS credentials, eliminating the need for long-lived AWS access keys in the CI/CD pipeline.
- Developed guardrailed AI-assisted remediation using runbook retrieval and risk-classified, allowlisted actions executed through SSM with human approval and recovery verification.
- Added **Prometheus metrics, health/readiness endpoints, CloudWatch monitoring, 10 incident runbooks, and Bash failure-injection scripts** for CPU, memory, disk, and service failures.

ExamPrep – AI Question Paper Generator GitHub

AWS EC2, Docker, GitHub Actions, GHCR, Nginx, React, Vite

- Containerized a React/Vite application using a **multi-stage Dockerfile**, producing an approximately **27 MB production image** served through Nginx on Ubuntu EC2.
- Built a GitHub Actions CI/CD pipeline that publishes Docker images to **GitHub Container Registry** using both **latest** and commit-SHA tags for version tracking and rollback.
- Automated EC2 deployment over SSH with container restart policies and post-deployment health checks, completing deployments in approximately one minute.
- Integrated **Gemini and Llama 3.3 70B through Groq** with provider failover for syllabus analysis and question generation.
- Implemented PDF and Word export functionality and managed deployment credentials through GitHub Secrets.
- Troubleshooted real deployment issues involving EC2 security groups, Docker permissions, and container replacement.

EDUCATION

C.K. College of Engineering and Technology, Cuddalore

Bachelor of Technology – Artificial Intelligence & Data Science

2026

CGPA: 8.5/10

Anandhan Memorial Matric Higher Secondary School

Higher Secondary (12th): 87.53% Secondary (10th): 84.6%